Kian Tomkins

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PROFILE

First-Class Software Engineering Master's graduate from Lancaster University, specialising in Embedded Systems, Distributed Systems and Human Computer Interaction. Proven track record in full-stack development, including building custom accessibility tools and abstracting low-level programming to high-level. Driven by improving people's lives with well-designed technology. Eager to bring strong technical fundamentals and a passion for software engineering into a collaborative development team.

WORK EXPERIENCE

Digital Development Industrial Experience | Hampshire Constabulary — Jun 2021 to Jun 2022

- Designed, developed, and maintained full-stack web and mobile applications.
- Migrated paper-based workflows into automated digital systems, including a room-booking mobile app, a temporary intranet login system, a uniform ordering site, and an automated meeting scheduler.
- Applied SOLID design to ensure maintainable architecture, alongside Git version control and Agile sprints.

Software Engineer Industrial Experience | Micro:bit Educational Foundation — Jan 2026 to Mar 2026

- Researched, designed, and developed a text-to-speech tool for the BBC micro:bit.
- Combined high-level web technology with low-level embedded systems under strict resource constraints.
- Produced detailed technical documentation to support decision-making and maintainability.

Student Software Engineer | Micro:bit Innovation and Research Lab — Jan 2026 to Jun 2026

- Led dissertation research that extended the work of the Micro:bit Educational Foundation.
- Improved the audio-playing capabilities of the micro:bit using additional hardware, to offset the limitations discovered in the MEWS prototype.
- Implemented low-level serial communication protocols, to ensure reliable audio storage and playback.
- Conducted thorough user studies with simulated impairments to accurately assess the impact of the Sound:Bit for those with Special Educational Needs.

Customer Service Manager | Entain — Sep 2022 to Oct 2026

- Ensured customer satisfaction and safeguarding, considering multiple stakeholders to resolve conflict.
- Ensured KPIs were met in a strictly-regulated industry.
- Aided customers in adjusting to a changing digital infrastructure.

EDUCATION & QUALIFICATIONS

Lancaster University — Oct 2022 to Jul 2026

- Master's Degree in Software Engineering (with Industrial Experience) – First Class Honours
- Psychology Minor – First Class

Fareham College — Sep 2020 to Jul 2022

- T-Level Digital Production Design and Development – Distinction

King's Academy Bay House **Sep 2015 to Jul 2020**

- 13 GCSEs, including Maths, English Language, English Literature and Computer Science.

KEY SKILLS:

- Languages: C, C++, C#, Python, Java, JavaScript, HTML, CSS, PHP, MySQL
- Frameworks: React, Bootstrap, Vue, PowerApps, Flask, .NET
- Developer Tools: Visual Studio, Android Studio, Git, Linux
- Additional Skills: REST API Development, Agile Development, SOLID Principles, Unit Testing

Technical Projects

Sound:Bit

Aims: To improve the audio-playing capabilities of the BBC Micro:bit, with additional hardware.

Hardware Design: A PCB with a SD card and a 3.5mm audio jack, allowing headphones or amplified speakers to be connected to the micro:bit.

Software Design: Files were generated by Kokoro.js, and loaded onto the micro:bit's SD card at compile time via WebUSB. The audio was then played-back using a double-buffered loop.

Technologies Used: C, EAGLE PCB Design, HTML, CSS Typescript, Bootstrap, REST APIs, Docker, PXT.

IoTA

Aims: To allow users to create routines and programmatically control IoT devices without technical logic.

Design: A drag-and-drop interface was created to allow the user to program the IoT devices in a similar fashion to Scratch or MakeCode. These blocks were then interpreted by a Python script that handled the networking and control between devices.

Technologies Used: Python, HTML, CSS, Typescript, Vue, MySQL, MQTT, Mosquitto, IoT, Docker.

Turing Texts (In Progress)

Aims: To create a game that assesses whether or not a person can distinguish between a Large Language Model and a person.

Design: A webpage with a C# backend relays messages from one user to either another user currently playing, using WebRTC; or a large language model connected through a REST API.

Technologies Used: C#, .NET, REST APIs, LLMs, Networking, HTML, CSS, Typescript, Docker.